



MPM3572

80V Input, 0.6A, Fully Integrated,
Synchronous, Step Down, Power Module

PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

DESCRIPTION

The MPM3572 is an 80V, 0.6A ultra-thin, fully integrated power module. It offers a full power solution that achieves up to 0.6A of output current with excellent load and line regulation over a wide input voltage range, from 10V up to 80V, and support both positive and negative output voltage from +/-3.3V up to +/-15V.

MPM3572 adopts constant-on-time (COT) control scheme, which provides ultra-fast transient response and simple loop compensation. The default operation frequency is 400kHz programmed by an internal resistor, and it is also optional that the operation frequency can be programmed easily from 400kHz to 1MHz by an external resistor.

MPM3572 features protection functions including over-current protection (OCP), and short-circuit protection (SCP), under voltage protection (UVP), and over-temperature protection (OTP).

MPM3572 requires a minimal number of readily available external components and is available in a LGA (6mmx6mmx2.0mm) package.

FEATURES

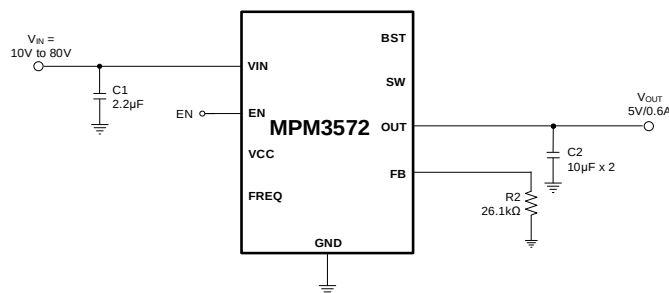
- 10V to 80V Input Voltage Range
- Positive and negative 3.3V to 15V output voltage range
- Up to 0.6A Output Current
- Low RDS(ON) Integrated Power MOSFETs
- Adaptive COT for Ultrafast Transient Response
- Programmable 400kHz to 1MHz frequency
- Internal Soft-Start and Loop Compensation
- OCP, SCP, UV, OTP with Hiccup
- High Efficiency PSM in Light Load
- Available in a LGA (6mmx6mmx2.0mm) Package

APPLICATIONS

- Industrial, Medical, Test Equipment
- GaN gate drive
- Sensors, Op-Amp Power

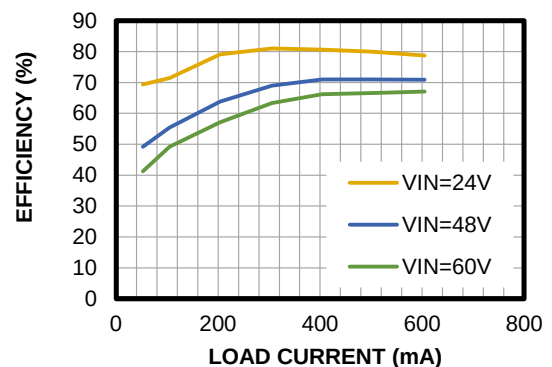
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TYPICAL APPLICATION



Efficiency vs. Load Current

$V_{OUT} = 5V$





ORDERING INFORMATION

Part Number*	Package	Top Marking	MSL Rating
MPM3572GMK	LGA (6.0mmx6.0mmx2.0mm)	See Below	3

* For Tray, add suffix -T (e.g. MPM3572GMK-T)

TOP MARKING

MPSYYWW

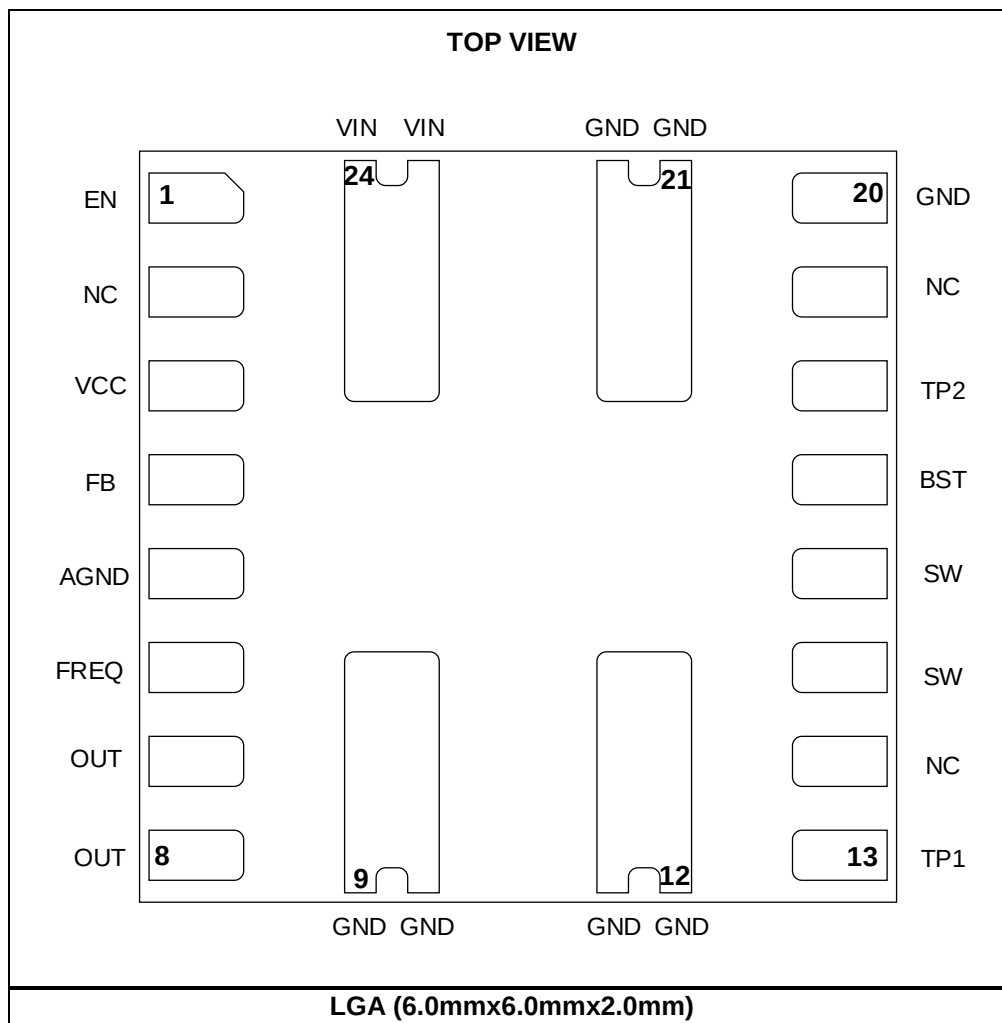
MP3572

LLLLLLLLLL

M

MPS: MPS prefix
YY: Year code
WW: Week code
MP3572: Part number
LLLLLLLLLL: Lot number
M: Module

PACKAGE REFERENCE





PIN FUNCTIONS

LGA 6X6 Pin #	Name	Description
1	EN	Power enable pin. Pull high to enable MPM3572, pull low to disable MPM3572. It cannot float, must be pulled high to enable the module or pulled low to disable the module.
2	NC	No Connection. Leave the pin floating, do not connect to any other pins.
3	VCC	Internal LDO output. Supply power to internal control circuit. A 1 μ F decoupling capacitor is connected internally between VCC and GND.
4	FB	VOUT voltage feedback pin. Connect a resistor between FB and GND to set the regulated output voltage. A 100kOhm resistor and a 1nF feed forward capacitor between FB and OUT are integrated inside the module.
5	AGND	Analog ground pin. Internally connected with GND.
6	FREQ	Switching frequency set pin. Default setting is 400kHz with internal 84.5kOhm resistor, an external resistor from this pin to GND sets the switching frequency up to 1MHz. Please find detail in OPERATION section.
7-8	OUT	Output of power module.
9-12, 20-22	GND	Power ground pin
13	TP1	Test pin. Leave the pin floating, do not connect to any other pins.
14	NC	No Connection. Leave the pin floating, do not connect to any other pins.
15-16	SW	Switch node of the power module. No connection needed.
17	BST	Bootstrap power pin for high side MOSFET Gate Driver. An internal capacitor is connected between BST and SW.
18	TP2	Test Pin. Leave the pin floating, do not connect to any other pins.
19	NC	No Connection. Leave the pin floating, do not connect to any other pins.
23-24	VIN	Power supply input pin.

ABSOLUTE MAXIMUM RATINGS ⁽¹⁾

Supply Voltage V_{IN} , EN	-0.3V to +90V
V_{SW}	-0.5V to +90V
$V_{SW}(<20ns)$	-0.5V to +96V
V_{BST} to SW	-0.3V to +6V
VOUT	-15V to +15V
All Other Pins	-0.3V to +6V
Continuous Power Dissipation ($T_A = +25^\circ C$) ⁽²⁾	
LGA (6mmx6mm)	2.87W
Junction Temperature	150°C
Lead Temperature	260°C
Storage Temperature	-65°C to +150°C

Recommended Operating Condition ⁽³⁾

Input Voltage Range	10V to 80V
Output Voltage	+/- 3.3V to 15V
Operating Junction Temp. (T_J)	-40°C to +125°C

Thermal Resistance ⁽⁴⁾

	θ_{JA}	θ_{JC}
LGA (6.0x6.0x2.0mm)	34.75	18.25. °C/W

NOTES:

- Exceeding these ratings may damage the device.
- The maximum allowable power dissipation is a function of the maximum junction temperature T_J (MAX), the junction-to-ambient thermal resistance θ_{JA} , and the ambient temperature T_A . The maximum allowable continuous power dissipation at any ambient temperature is calculated by $P_D (MAX) = (T_J (MAX) - T_A) / \theta_{JA}$. Exceeding the maximum allowable power dissipation produces an excessive die temperature.
- The device is not guaranteed to function outside of its operating conditions.
- Measured on EVM3572-MK-00A, 4-layer PCB.

**ELECTRICAL CHARACTERISTICS**

$V_{IN} = 48V$, $T_J = -40^{\circ}C$ to $+125^{\circ}C$ ⁽⁵⁾, typical value is tested at $T_J = 25^{\circ}C$, unless otherwise noted.

Parameters	Symbol	Condition	Min	Typ	Max	Units
Power Supply						
VCC Under Voltage Lockout Threshold Rising	$V_{CCUVLO-R}$	VCC Rising	3.15	3.35	3.55	V
VCC Under Voltage Lockout Threshold Hysteresis	V_{CCHYS}			150		mV
VCC Regulation Voltage		$I_{VCC}=2mA$		4.85	5.2	V
Shutdown Current	I_{SD}	$V_{EN}=0V$, Measured on VIN pin, $T_J=25^{\circ}C$			2	μA
Quiescent Current	I_Q	$V_{IN}=12V$, $V_{FB}=1.05V$, no switching, measured on VIN pin, $T_J=25^{\circ}C$		15		μA
Enable Control						
Micro Power EN Startup Threshold			0.9			V
Micro Power EN Off Threshold					0.4	V
EN Input High Threshold	V_{EN-ON}	V_{EN} Rising	1.15	1.25	1.35	V
EN Hysteresis	V_{EN-H}	V_{EN} Rising/Falling		120		mV
Frequency						
Minimum Off Time ⁽⁶⁾	$T_{MIN-OFF}$	$V_{FB}=0V$		150		ns
Minimum On Time ⁽⁶⁾	T_{MIN-ON}			45		ns
Reference Voltage						
FB Reference Voltage	V_{REF}	$V_{IN}=8V$ to $80V$, $T_J=25^{\circ}C$	0.99	1	1.01	V
FB Input Current	I_{FB}	$V_{FB}=1.05V$			50	nA
Soft-Start Time	t_{SS}	10% to 90% of V_{REF} , $T_J=25^{\circ}C$		4		ms
FB Under Voltage Threshold	V_{UVP}			60%		V_{REF}
Power Switch						
Low-Side Switch On-Resistance	R_{ON-L}			265		m Ω
High-Side Synchronous Switch On-Resistance	R_{ON-H}			625		m Ω



ELECTRICAL CHARACTERISTICS

$V_{IN} = 48V$, $T_J = -40^{\circ}C$ to $+125^{\circ}C$ ⁽⁵⁾, typical value is tested at $T_J = 25^{\circ}C$, unless otherwise noted.

Parameters	Symbol	Condition	Min	Typ	Max	Units
Current Limit						
Low Side FET Valley Current Limit	$I_{VL-LIMIT}$			0.63		A
Inductor Current zero current detection (ZCD) Threshold			-100	0	100	mA
Thermal Protection						
Thermal shutdown ⁽⁶⁾	T_{SD}			150		$^{\circ}C$
Thermal shutdown Hysteresis ⁽⁶⁾	T_{SD-HYS}			25		$^{\circ}C$

Notes:

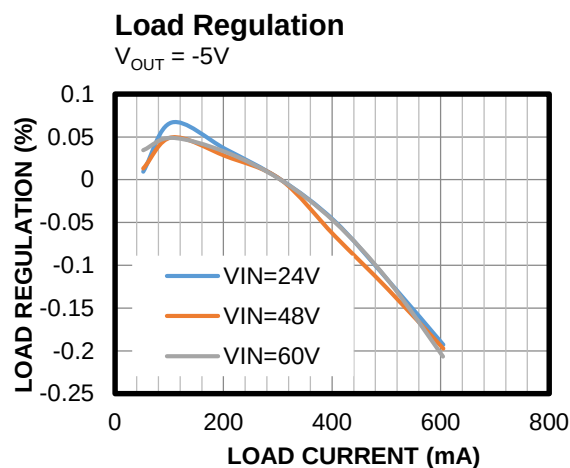
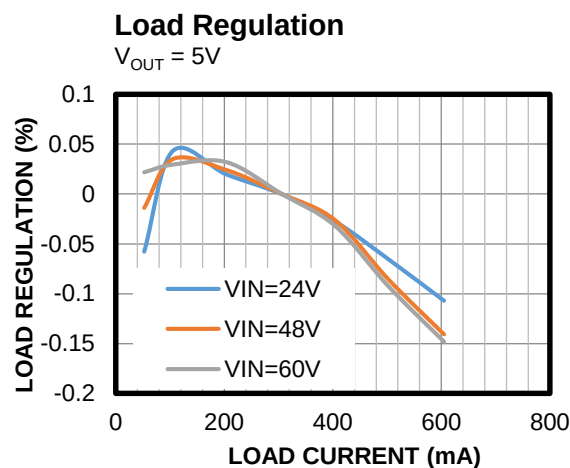
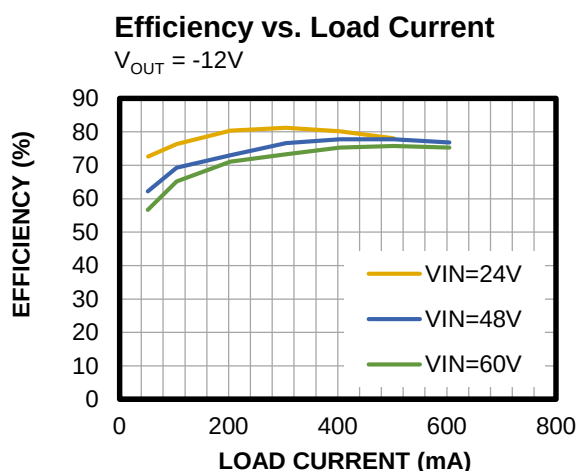
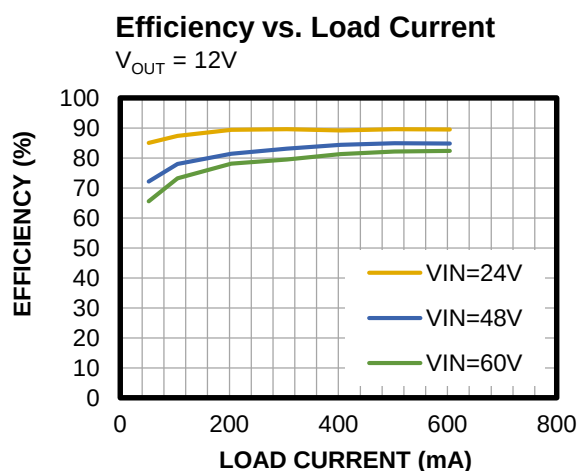
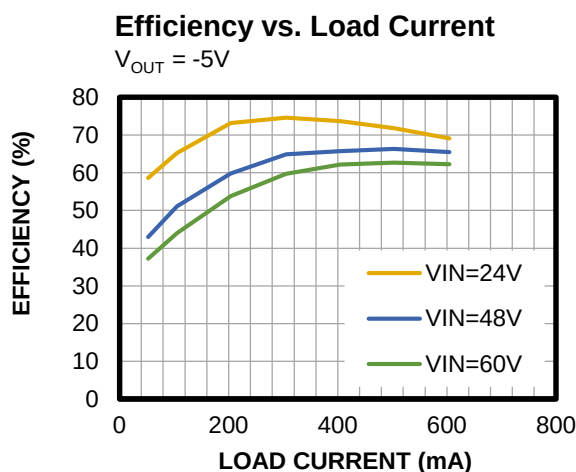
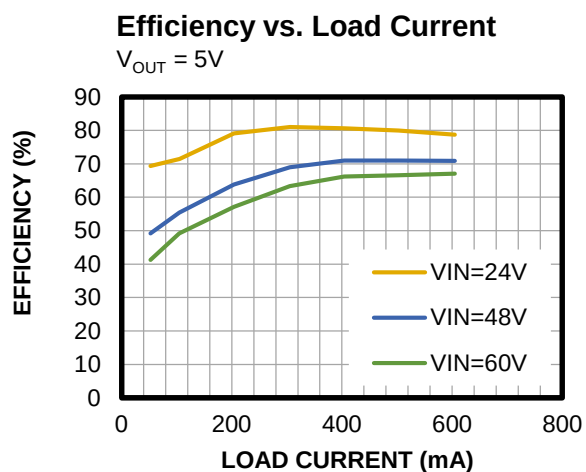
5) Not tested in production; guaranteed by over-temperature correlation

6) Guaranteed by engineering sample characterization.



TYPICAL PERFORMANCE CHARACTERISTICS

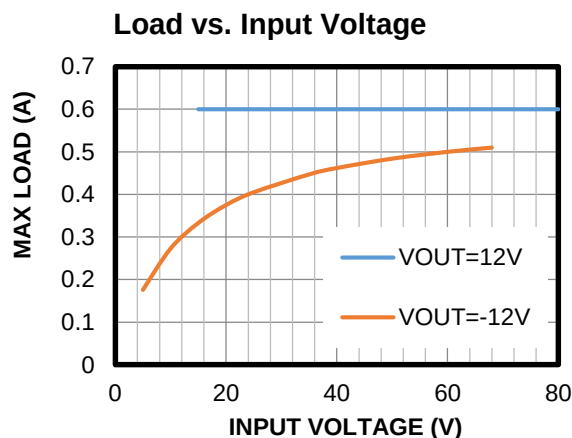
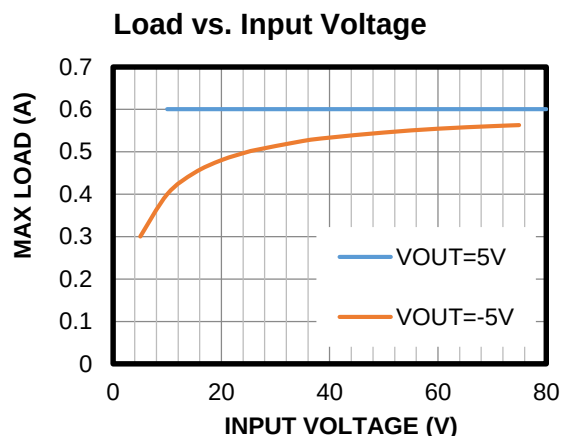
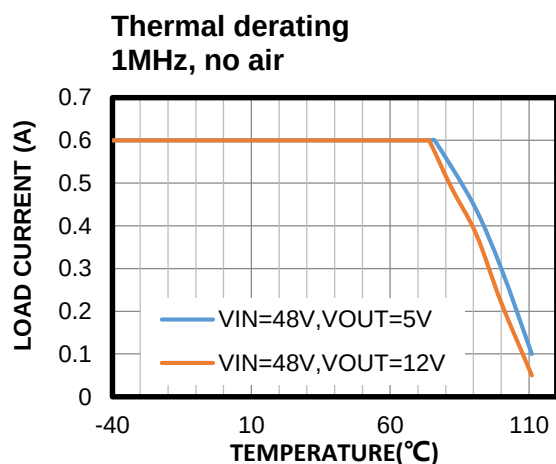
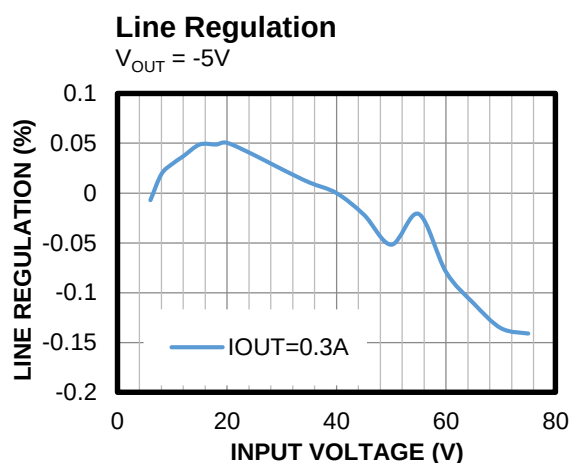
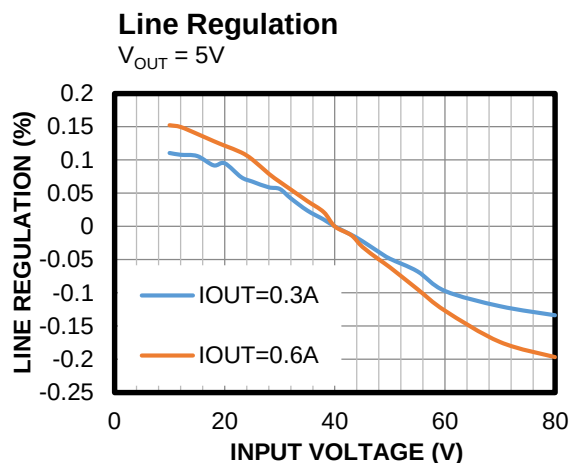
$V_{IN} = 48V$, $V_{OUT} = 5V$, $T_A = 25^\circ C$, unless otherwise noted.





TYPICAL PERFORMANCE CHARACTERISTICS (continued)

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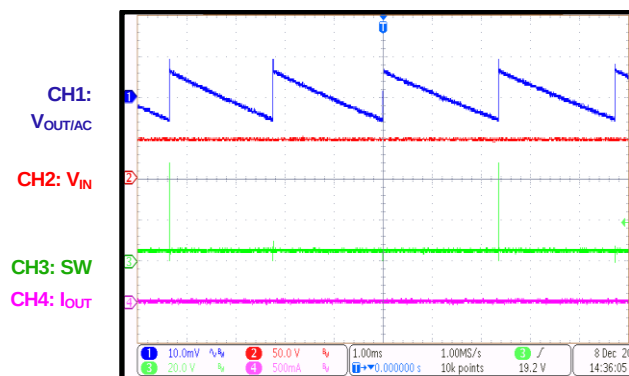




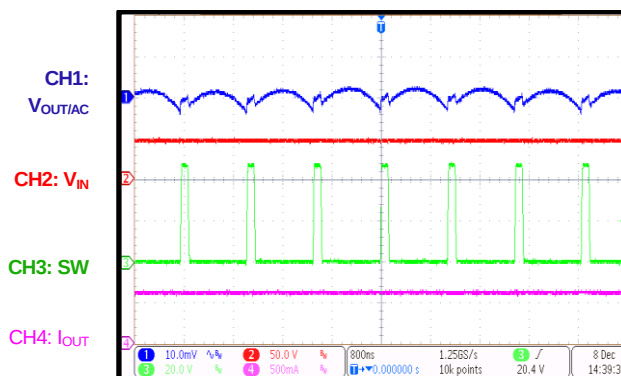
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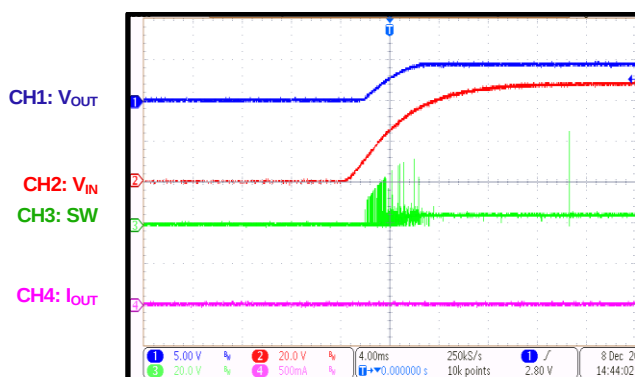
Steady State

 $V_{IN} = 48V$, $V_{OUT} = 5V$, $f_{SW} = 1000kHz$, $I_{OUT} = 0A$


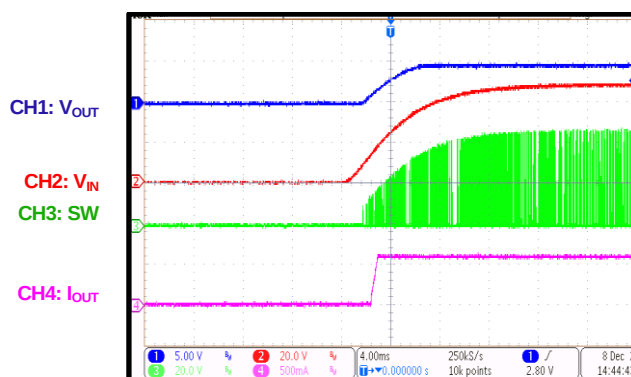
Steady State

 $V_{IN} = 48V$, $V_{OUT} = 5V$, $f_{SW} = 1000kHz$, $I_{OUT} = 0.6A$


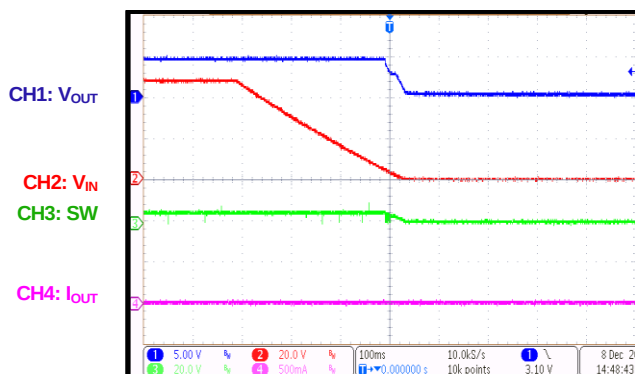
VIN ON

 $V_{IN} = 48V$, $V_{OUT} = 5V$, $f_{SW} = 1000kHz$, $I_{OUT} = 0A$


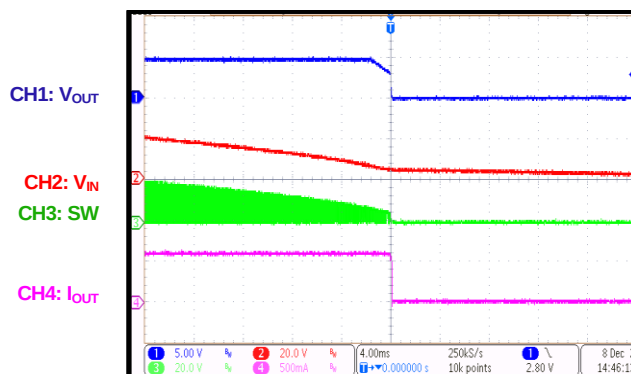
VIN ON

 $V_{IN} = 48V$, $V_{OUT} = 5V$, $f_{SW} = 1000kHz$, $I_{OUT} = 0.6A$


VIN OFF

 $V_{IN} = 48V$, $V_{OUT} = 5V$, $f_{SW} = 1000kHz$, $I_{OUT} = 0A$


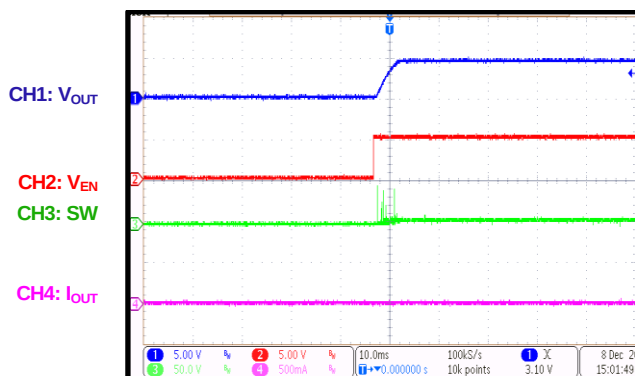
VIN OFF

 $V_{IN} = 48V$, $V_{OUT} = 5V$, $f_{SW} = 1000kHz$, $I_{OUT} = 0.6A$


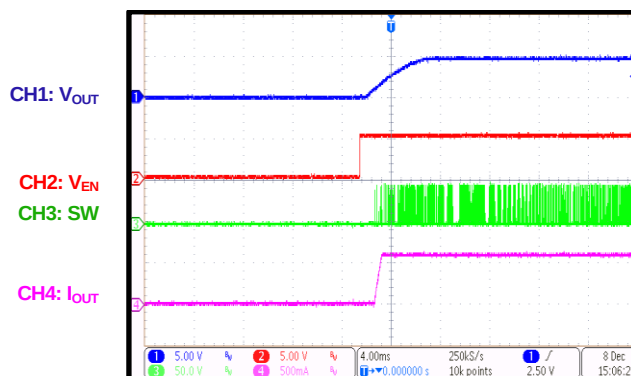
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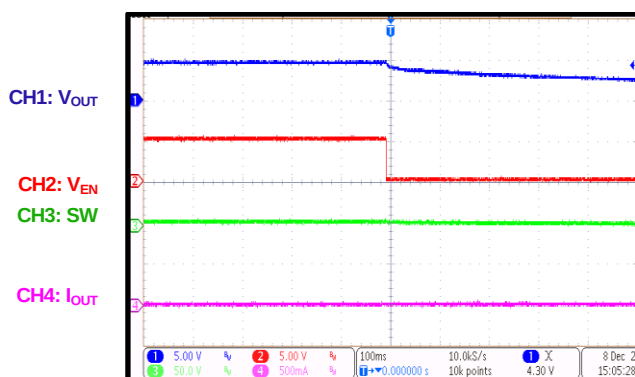
EN ON

 $V_{IN} = 48V$, $V_{OUT} = 5V$, $f_{SW} = 1000kHz$, $I_{OUT} = 0A$ 

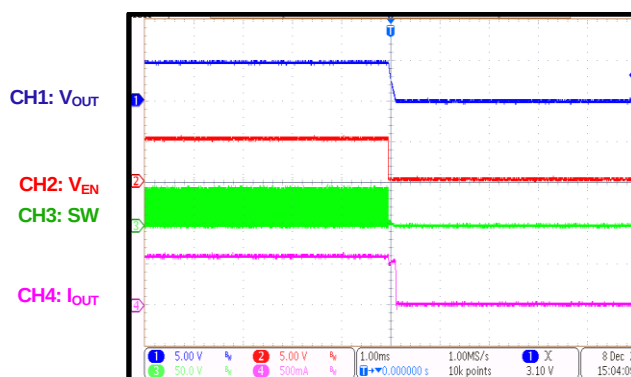
EN ON

 $V_{IN} = 48V$, $V_{OUT} = 5V$, $f_{SW} = 1000kHz$, $I_{OUT} = 0.6A$ 

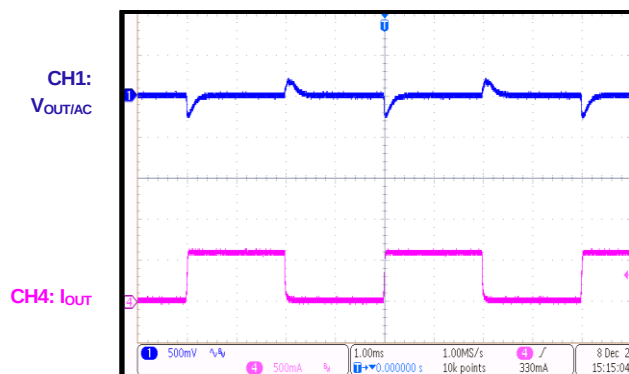
EN OFF

 $V_{IN} = 48V$, $V_{OUT} = 5V$, $f_{SW} = 1000kHz$, $I_{OUT} = 0A$ 

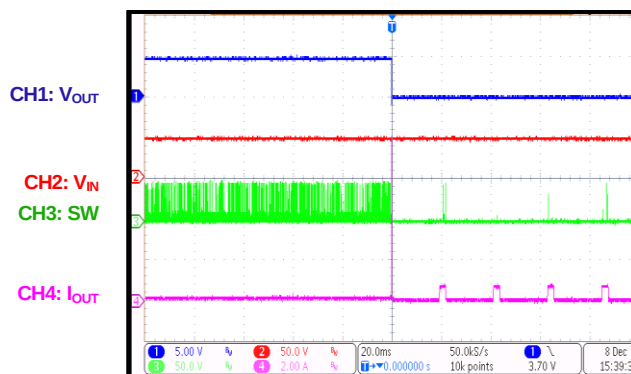
EN OFF

 $V_{IN} = 48V$, $V_{OUT} = 5V$, $f_{SW} = 1000kHz$, $I_{OUT} = 0.6A$ 

Load Transient

 $V_{IN} = 48V$, $V_{OUT} = 5V$, $I_{OUT} = 0-0.6A$, $0.8A/\mu s$ 

SCP Enter

 $V_{IN} = 48V$, $V_{OUT} = 5V$, $f_{SW} = 1000kHz$, $I_{OUT} = 0A$ 

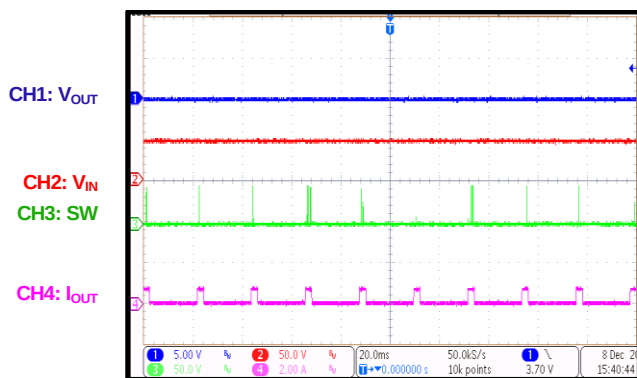


TYPICAL PERFORMANCE CHARACTERISTICS (continued)

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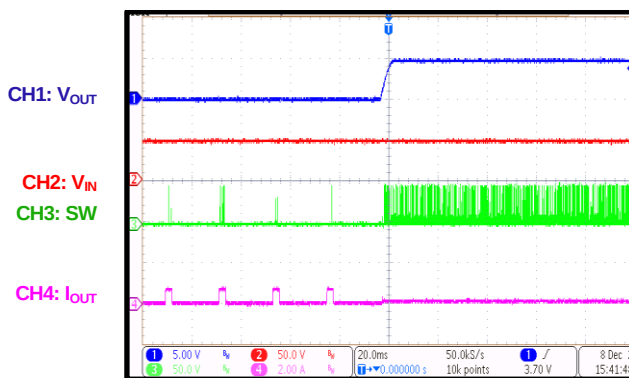
SCP Steady State

$V_{IN} = 48V$, $V_{OUT} = 5V$, $f_{SW} = 1000kHz$, $I_{OUT} = 0A$



SCP Recovery

$V_{IN} = 48V$, $V_{OUT} = 5V$, $f_{SW} = 1000kHz$, $I_{OUT} = 0A$





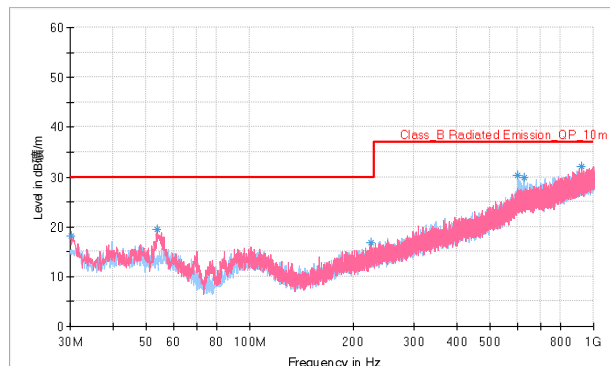
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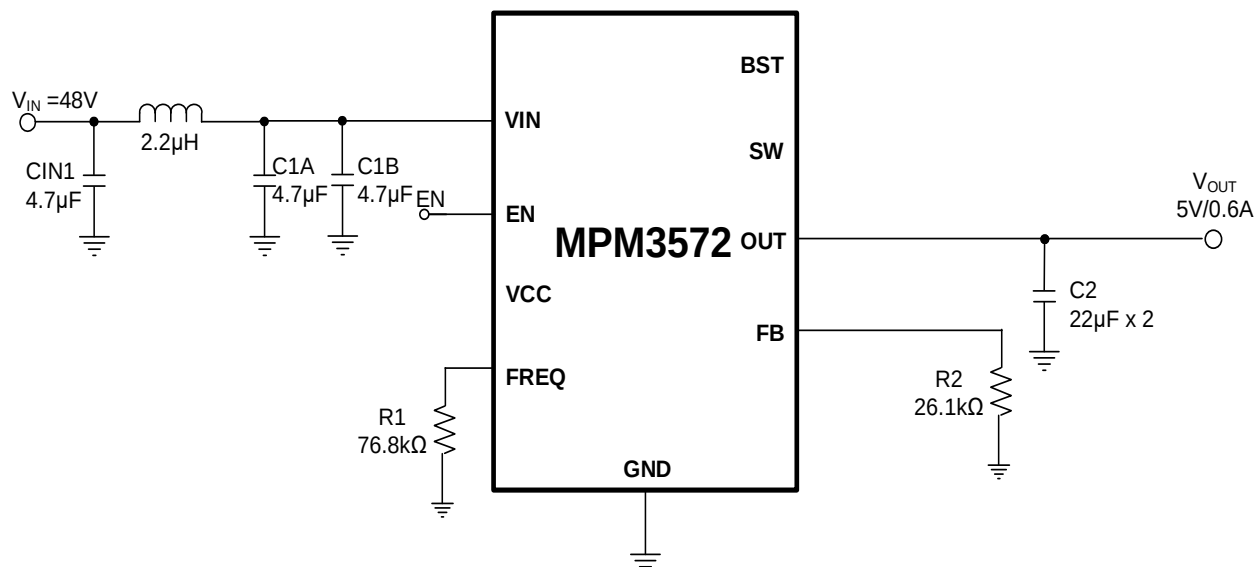
Radiated Emission, EN55022 Class B

MPM3572, $V_{in}=48V$, $V_{out}=5V$

EMI filter: $4.7\mu F$, $2.2\mu H$



EMI TEST CIRCUIT



BLOCK DIAGRAM

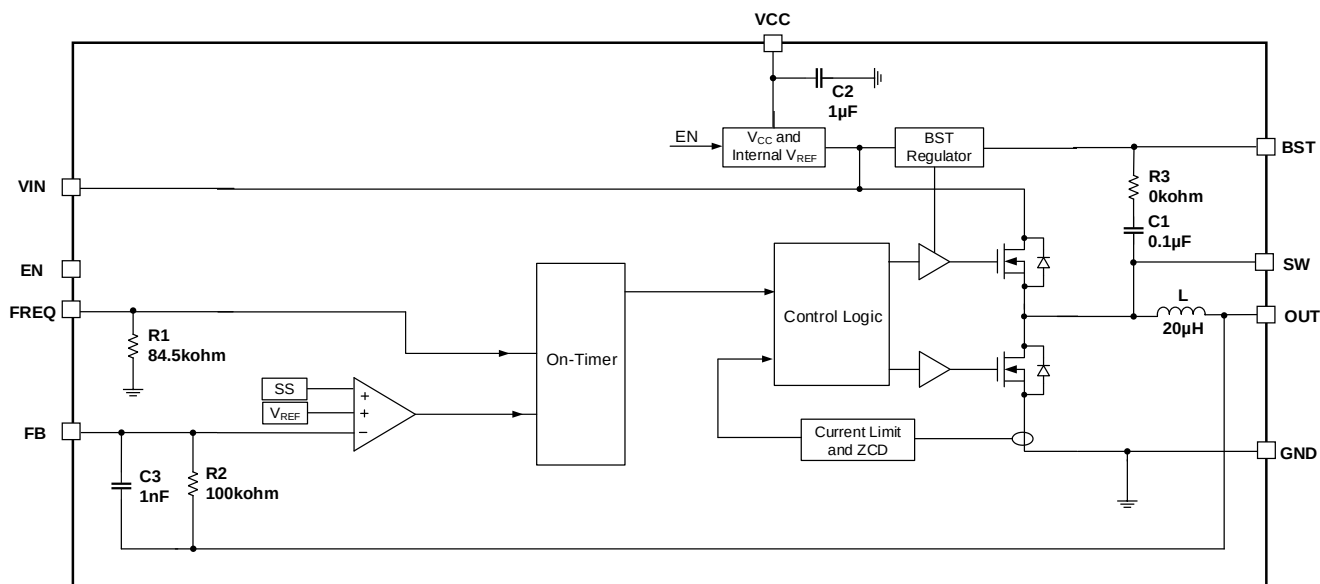


Figure 1: Functional Block Diagram

OPERATION

MPM3572 is a fully integrated high voltage power module with wide input voltage range from 10V to 80V, and it is capable of both positive and negative output voltage range from +/- 3.3V to +/-15V. MPM3572 delivers up to 0.6A continuous output current on a 6mmx6mmx2.0 mm ultra-thin LGA package.

MPM3572 provides a highly efficient solution with COT control for very fast loop response, and the control scheme eases loop stabilization. It features internal soft-start control, precise current limiting. The ultra-low quiescent current (I_Q) makes it suitable for light load high efficiency applications.

COT Operation

MPM3572 works with constant-on-time (COT) control scheme. At the beginning of each cycle, the high-side MOSFET (HS-FET) is turned ON for a fixed on time when the FB voltage drops below reference voltage. The fixed on time is determined by one-shot on-timer. The on time is adaptive since it depends on both of the output voltage and input voltage to make the switching frequency fairly constant over wide input voltage range.

After the ON period ends, the HS-FET is turned off until next period. Through this way, the regulation of output voltage is achieved.

To minimize the conduction loss, the low-side MOSFET (LS-FET) works as synchronized rectifier. A dead time is internally generated and inserted, during the transient when one of the switch is turning on and the other is turning off, to avoid shoot-through.

Heavy-Load Operation

When the output current is high and the inductor current is always above zero amps, MPM3572 is working in the continuous-conduction-mode (CCM). In the CCM mode, HS-FET is turned on at the beginning of each switching cycle, and turned off after the ON period. Then the LS-FET is turned on to conduct the inductor current.

During the CCM operation, the switching frequency is relatively constant and it is called PWM mode.

Light-load operation

MPM3572 works under PSM in the light load condition. In the PSM operation, the LS-FET driver goes into tri-state (high-Z) when inductor current drop near zero, at the same time, the output capacitors discharge slowly to GND through FB feedback resistor. If the VOUT drops and internal COMP rises, MPM3572 starts the next switching cycle by turning on HS-FET again. MPM3572 automatically reduces the switching frequency and reduce the quiescent current when the IC is not switching. This operation significantly improves device efficiency when the output current is low.

In light-load PSM operation, the HS-FET does not turn on with same frequency as that during heavy load operation. The actual frequency is the function of the output current. As the output current increases, HS-FET operates with higher frequency accordingly. As a result, f_{sw} also increases. I_{OUT} reaches its critical level when the valley inductor current reaches 0A. I_{OUT} can be calculated with equation (1):

$$I_{OUT} = \frac{(V_{IN} - V_{OUT}) \times V_{OUT}}{2 \times L \times f_{SW} \times V_{IN}} \quad (1)$$

where f_{sw} is the switching frequency.

MPM3572 reverts to CCM operation once I_{OUT} exceed the critical level. After this point, f_{sw} remains constant across the entire I_{OUT} range.

Switching frequency (f_{sw})

Selecting a switching frequency (f_{sw}) is a tradeoff between efficiency and component size. Low-frequency operation increases efficiency by reducing MOSFET switching loss, but requires a larger inductance and capacitance to maintain the low output voltage ripple.

The MPM3572 offers configurable COT control. When the FREQ pin is connected to the GND pin via a resistor (R_{FREQ}), V_{IN} is internally fed-forward to the one-shot on-timer. When the MPM3572 is in steady-state operation in CCM, the duty ratio is set as V_{OUT} / V_{IN} . Therefore, f_{sw} remains constant across the entire input voltage range. f_{sw} can be estimated with Equation (2):



$$f_{sw} = \left[\frac{10^6 \times (84.5 + R_{FREQ}(k\Omega))}{34.5 \times R_{FREQ}(k\Omega) \times 84.5} + 57 \right] (\text{kHz}) \quad (2)$$

By adjusting the external R_{FREQ} , the switching frequency can range from 400kHz up to 1MHz.

Power Supply

MPM3572 features an LDO for internal control circuit, and it is regulated by VIN. The output of LDO can be measured by VCC pin.

The internal VCC circuit also features a VCC UVLO circuit to protect the entire module from operating at insufficient supply voltage. The voltage of VCC is monitored and the module only starts to work when VCC exceeds the VCC UVLO rising threshold.

Start Up

When EN is high and VCC is higher than UVLO, MPM3572 start up with internal Soft-Start signal. After MPM3572 starts, SS signal ramps up from 0V and compared with V_{REF} , the lower voltage will feed to the error amplifier (EA) to control output voltage. After SS signal rises to above V_{REF} , soft-start completes and internal reference takes charge of the feedback loop regulation.

MPM3572 has been designed for monotonic startup into pre-bias loads. If the output is pre-biased to a certain voltage during startup, the voltage on the soft-start capacitor will be charged. If the soft-start capacitor voltage exceeds the sensed output voltage at the FB, the module starts to work normally.

Enable (EN) and Configurable Under-Voltage Lockout (UVLO)

The EN pin turns the MPM3572 on and off. If voltage above 0.9V is applied, the MPM3572 starts the internal bandgap (BG) circuit. Once the EN voltage (V_{EN}) exceeds EN's high threshold (typically 1.25V), the MPM3572 turns on all functions and starts switching operation. Switching operation stops once V_{EN} drops below the low threshold (typically 1.13V). EN is compatible with voltages up to 90V.

There is no internal resistor on EN. For automatic start-up, connect EN to VIN through a resistor.

Over Current and Short Circuit Protection

MPM3572 features valley current limit control. During HS-FET is in ON state, the inductor current is monitored. When the sensed inductor current is higher than the valley current limit, the LS limit comparator turns over, HS-FET will wait until inductor current falls below valley current limit to turn on again. Meanwhile, the output voltage (VFB) may drop below the under voltage threshold, which is typically 60% of the reference voltage. Once UVLO is triggered, the MPM3572 enters hiccup mode to periodically restart. During soft-start period, the hiccup protection is disabled.

During over-current hiccup protection, MPM3572 disables output power stage, discharge the SS capacitor. Then MPM3572 automatically initiates another soft start. If the over-current condition still exists after soft-start ends, the module repeats the hiccup operation until over-current conditions disappear and then V_{OUT} rises back to regulation level.

BST Power Supply

An internal bootstrap capacitor powers the floating HS-FET driver. The bootstrap capacitor is charged by VCC. The $V_{BST}-V_{SW}$ voltage can supply the power of HS-FET driver.

If both VCC and EN exceed the respective thresholds, the module starts. The reference blocks start first, generating stable reference voltage and reference currents, and then the internal regulator is enabled. The regulator provides a stable supply for the remaining circuits.

Three events can shut down the module: EN low, VCC low, and thermal shutdown. The shutdown procedure starts by initially blocking the signaling path to avoid any fault triggering. The COMP voltage and the internal supply rail are then pulled down. The BST floating driver is not subject to this shutdown command.

Thermal Protection

MPM3572 integrates a temperature monitor circuits. Once junction temperature is higher than 150°C, MPM3572 shuts down. After the temperature drops below 125°C, the power supply resumes operation.

APPLICATION INFORMATION

COMPONENT SELECTION

Setting the Output Voltage

The voltage divider for feedback output voltage is partially integrated inside the module. A 100kOhm pull up resistor R1 together with a 1nF capacitor in parallel is integrated inside the module. A resistor R2 is required externally to complete the entire feedback resistor divider, based on the regulated voltage level.

For MPM3572, R2 need to be selected based on the regulated voltage level and internal 100kOhm pull up resistor. A smaller R2 will lead to considerable quiescent current loss while too large R2 makes the FB noise sensitive. It can be determined by Equation (3):

$$R2(k\Omega) = \frac{100(k\Omega) \times V_{REF}}{V_{OUT} - V_{REF}} \quad (3)$$

Where V_{REF} is the reference voltage, and it is 1.0V typically. The output voltage feedback gain is determined by Equation (4):

$$G_{FB} = \frac{R2(k\Omega)}{100 + R2(k\Omega)} \quad (4)$$

To optimize the load transient response, a 1nF feed-forward capacitor (C_{FF}) is integrated inside of the module in parallel with R1. Recommended divider resistors and frequency as below table 1 shows:

Table1. Recommended divider resistors and frequency

Work condition	R2	Frequency
VOUT=5V	26.1k	400kHz~1000kHz
VOUT=-5V	26.1k	400kHz~1000kHz
VOUT=12V	9.1k	650kHz~1000kHz
VOUT=-12V	9.1k	650kHz~1000kHz

Selecting Input Capacitor

The input current to a buck converter is discontinuous, and therefore, requires a capacitor to supply the AC current to the step-down module while maintaining the DC input voltage. Use ceramic capacitors for the best

performance. During layout, place the input capacitors as close to the VIN pin as possible.

The capacitance can vary significantly with temperature. Use capacitors with X5R and X7R ceramic dielectrics because they are fairly stable over a wide temperature range.

The capacitors must also have a ripple current rating that exceeds the converter's maximum input ripple current. Estimate the input ripple current by Equation (5):

$$I_{CIN} = I_{OUT} \times \sqrt{\frac{V_{OUT}}{V_{IN}} \times \left(1 - \frac{V_{OUT}}{V_{IN}}\right)} \quad (5)$$

The worst-case condition occurs at $V_{IN} = 2V_{OUT}$, calculated with Equation (6):

$$I_{CIN} = \frac{I_{OUT}}{2} \quad (6)$$

For simplification, choose an input capacitor with an RMS current rating that exceeds half the maximum load current.

The input capacitance value determines the converter input voltage ripple. Select a capacitor value that meets any input voltage ripple requirements.

Estimate the input voltage ripple by Equation (7):

$$\Delta V_{IN} = \frac{I_{OUT}}{F_{SW} \times C_{IN}} \times \frac{V_{OUT}}{V_{IN}} \times \left(1 - \frac{V_{OUT}}{V_{IN}}\right) \quad (7)$$

The worst-case condition occurs at $V_{IN} = 2V_{OUT}$, can be calculated by Equation (8):

$$\Delta V_{IN} = \frac{1}{4} \times \frac{I_{OUT}}{F_{SW} \times C_{IN}} \quad (8)$$

Selecting Output Capacitor

The output capacitor maintains the DC output voltage. Use ceramic capacitors or POSCAPs. Estimate the output voltage ripple by Equation (9):

$$\Delta V_{OUT} = \frac{V_{OUT}}{F_{SW} \times L} \times \left(1 - \frac{V_{OUT}}{V_{IN}}\right) \times \left(R_{ESR} + \frac{1}{8 \times F_{SW} \times C_{OUT}}\right) \quad (9)$$

When using ceramic capacitors, the capacitance dominates the impedance at the switching frequency. The capacitance also dominates the



output voltage ripple. For simplification, estimate the output voltage ripple by Equation (10):

$$\Delta V_{OUT} = \frac{V_{OUT}}{8 \times F_{SW}^2 \times L \times C_{OUT}} \times \left(1 - \frac{V_{OUT}}{V_{IN}}\right) \quad (10)$$

With POSCAP capacitors, the ESR dominates the impedance at the switching frequency. For simplification, the output ripple can be estimated with Equation (11):

$$\Delta V_{OUT} = \frac{V_{OUT}}{f_{SW} \times L} \times \left(1 - \frac{V_{OUT}}{V_{IN}}\right) \times R_{ESR} \quad (11)$$

Selecting a larger output capacitor provides faster load transient response. However, the maximum output capacitor limit should be also considered when designing for application. If the output capacitor value is too high, the output voltage cannot reach the desired value during the SS time, and the capacitor fails to regulate. The maximum output capacitor value (C_{O_MAX}) can be estimated with Equation (12):

$$C_{O_MAX} = (I_{LIM_AVG} - I_{OUT}) \times t_{SS} / V_{OUT} \quad (12)$$

Where I_{LIM_AVG} is the average start-up current during the soft-start period, and t_{SS} is the SS time.

PCB Layout Guidelines

An optimized PCB layout is critical for reliable operation. For best result, refer to Figure 2 and follow the below guidelines below:

1. Place the input MLCC capacitors as close to the VIN and GND pins as possible.
2. Maximize the VIN and GND copper plane to minimize parasitic impedance.
3. Ensure that the high-current paths (GND, VIN, and OUT) have short and wide traces.
4. Place as many GND vias as possible close to the pin to minimize parasitic impedance and thermal resistance.
5. Place the external feedback resistor next to FB.

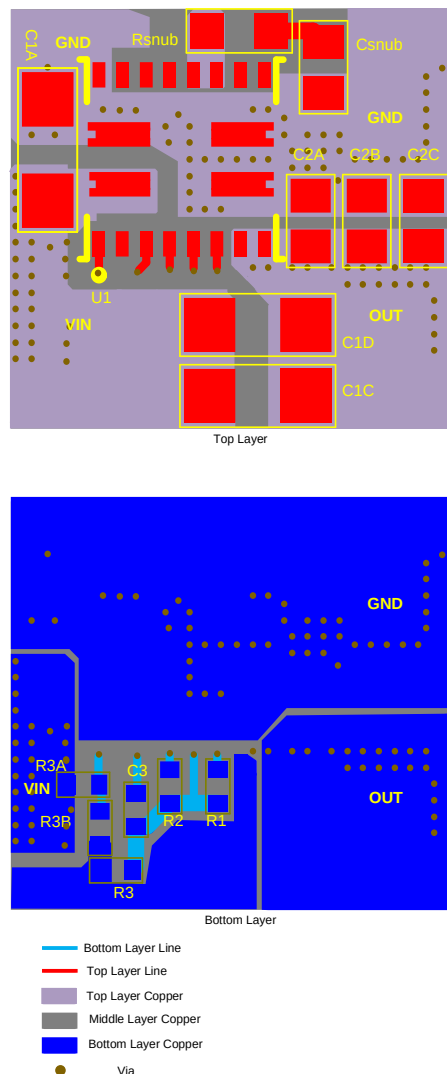


Figure 2: Recommended PCB Layout

Note:

- 7) This layout covers full range of specs. Much smaller layout size can be achieved for specific cases like higher switching frequency, lower input voltage or output current by selecting smaller packages of capacitors.

TYPICAL APPLICATION CIRCUITS

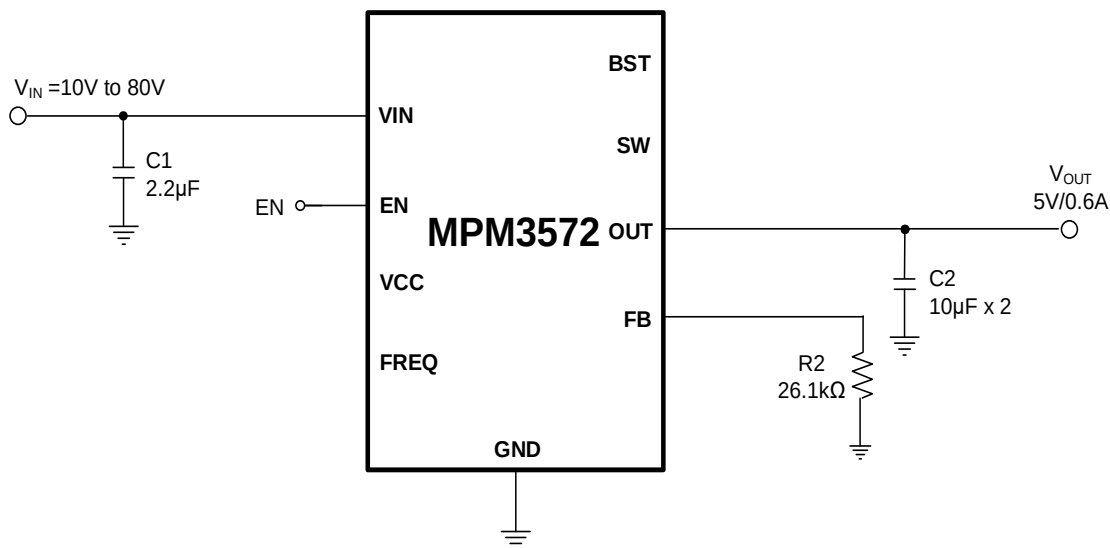


Figure 3: Typical application with external EN control and positive output

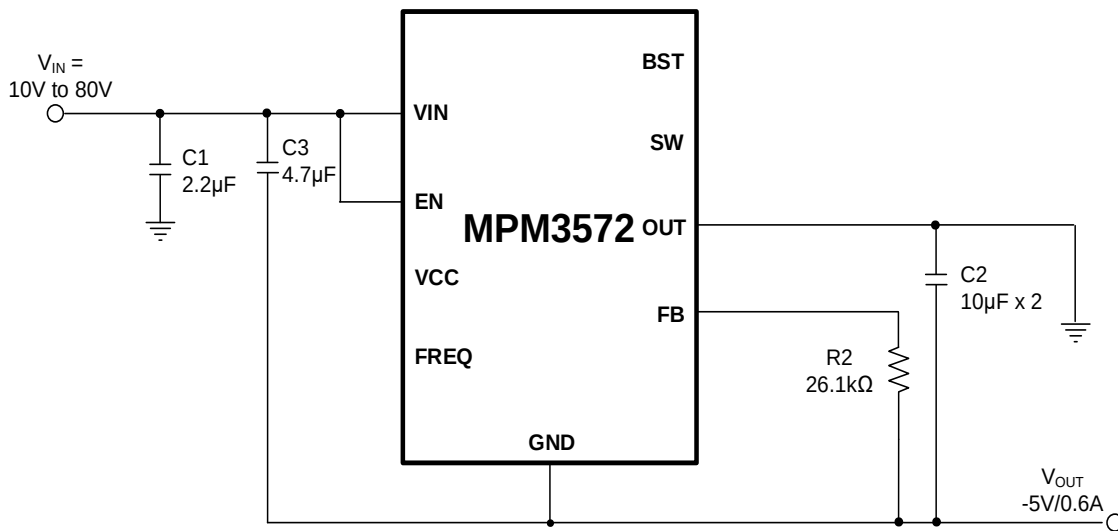
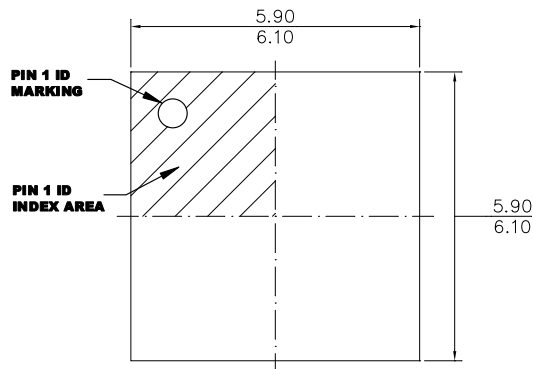


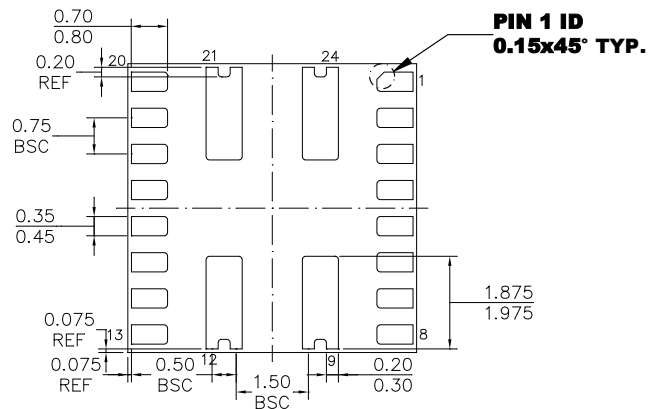
Figure 4: Typical application with automatic start up and negative output

PACKAGE INFORMATION

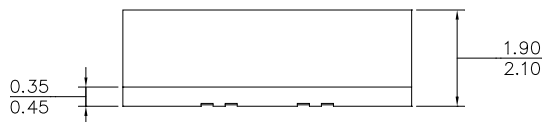
LGA (6.0mmx6.0mmx2.0mm)



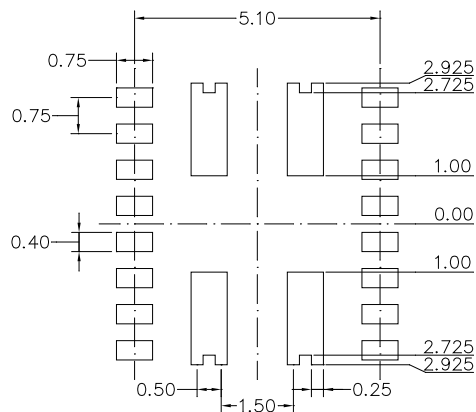
TOP VIEW



BOTTOM VIEW



SIDE VIEW

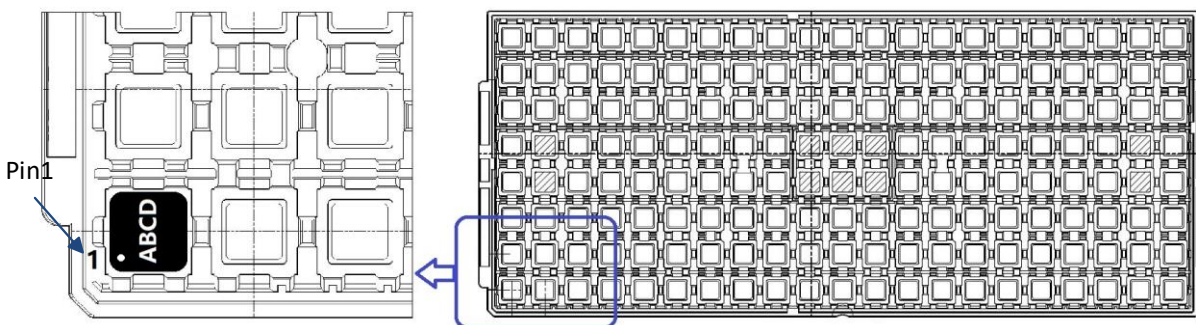


RECOMMENDED LAND PATTERN

NOTE:

- 1) ALL DIMENSIONS ARE IN MILLIMETERS.
- 2) LEAD COPLANARITY SHALL BE 0.10 MILLIMETERS MAX.
- 3) JEDEC REFERENCE IS MO-303.
- 4) DRAWING IS NOT TO SCALE.

CARRIER INFORMATION



Part Number	Package Description	Quantity /Reel	Quantity /Tube	Quantity /Tray	Reel Diameter	Carrier Tape Width	Carrier Tape Pitch
MPM3572GMK-T	MPM LGA6x6x2.0	N/A	N/A	490	N/A	N/A	N/A

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